(1)

$P(T) = 1 - P(\bar{T})$

$= 1 - P(\bar{T}_1 \cap \bar{T}_2 \cap \bar{T}_3 \cap \bar{T}_4)$

$= 1 - (1-\frac{1}{4})\cdot(1-\frac{1}{2})(1-\frac{1}{3})(1-\frac{1}{2})$

$= 1 - \frac{\cancel{3}}{4}\cdot\frac{1}{2}\cdot\frac{\cancel{2}}{\cancel{3}}\cdot\frac{1}{2}$

$= 1 - \frac{1}{8}$

$= \frac{7}{8}$

(2) $P(D) = P(B \cap C)$

$= p_b \cdot p_c$

$P(S) = P(A \cup D)$

$= p_a + p_b p_c - p_a p_b p_c$

(3) (b) $T \in [1, \infty)$

$a \cdot \int_1^\infty \frac{1}{t^4}\, dt \overset{!}{=} 1$

$= a\cdot\left(\frac{1}{-3t^3}\right)\Big|_1^\infty$

$= a\cdot\left(0 - \frac{1}{-3}\right)$

$= a \cdot \frac{1}{3} \neq 1$

$= 1$

$3 = a$

$f_T(t) = \begin{cases} 3\cdot\frac{1}{t^4} & , t \geq 1 \\ 0 & , t < 1 \end{cases}$

(c) $F_T(t) = \int_{-\infty}^{t} f_T(\tilde{t})\, d\tilde{t}$

$= \int_{-\infty}^{1} f_T(\tilde{t})\, d\tilde{t} + \int_1^t f_T(\tilde{t})\, d\tilde{t}$

$= 0 + \int_1^t 3\frac{1}{\tilde{t}^4}\, d\tilde{t}$

$= \frac{\cancel{3}}{\cancel{3}} \frac{1}{-t^3}\Big|_1^t$

$= \frac{1}{-t^3} - \frac{1}{-1^3}$

$= 1 - \frac{1}{t^3}$

(d) Median, $Q_{1/4}$, $Q_{3/4}$

$F_T(t) = \begin{cases} 1 - \frac{1}{t^3} & , t \geq 1 \\ 0 & , t < 1 \end{cases}$

Median:

$1 - \frac{1}{m^3} = \frac{1}{2}$

$1 - \frac{1}{2} = \frac{1}{m^3}$

$\frac{1}{2} = \frac{1}{m^3}$

$m^3 = 2$

$m = \sqrt[3]{2} = 1{,}26$

$Q_{1/4}$:

$1 - \frac{1}{Q_{1/4}^3} = \frac{1}{4}$

$1 - \frac{1}{4} = \frac{1}{Q_{1/4}^3}$

$\frac{3}{4} = \frac{1}{Q_{1/4}^3}$

$Q_{1/4}^3 = \frac{4}{3}$

$Q_{1/4} = \sqrt[3]{\frac{4}{3}} = 1{,}10$

$Q_{3/4}$:

$1 - \frac{1}{Q_{3/4}^3} = \frac{3}{4}$

$1 - \frac{3}{4} = \frac{1}{Q_{3/4}^3}$

$\frac{1}{4} = \frac{1}{Q_{3/4}^3}$

$Q_{3/4} = \sqrt[3]{4} = 1{,}587$

$Q_p \to$ Interquartilsabstand

$Q_p = Q_{3/4} - Q_{1/4} = 1{,}587 - 1{,}1$

$= 0{,}487$

(4)

$\lambda = \frac{1}{8\,a}$

$t = 2a$

(a)

$$F_T(t) = 1 - e^{-\lambda \cdot t}$$
$$= 1 - e^{-\frac{1}{8a} \cdot 2a}$$
$$= 1 - e^{-\frac{1}{4}} = 1 - 0{,}779 = 0{,}221$$

(b)

$$E[T] = \frac{1}{\lambda} = \frac{1}{\frac{1}{8a}} = \underline{8\,a}$$

$$Var[T] = \frac{1}{\lambda^2} = \frac{1}{\frac{1}{64a^2}} = \underline{64\,a^2}$$

(6)

Gleichverteilung, Binomialverteilung, Poisson-Verteilung
Hypergeometrische Verteilung, Geometrische Verteilung, Bernoulli Verteilung

(7)

Gleichverteilung, Exponentialverteilung, Gamma-Verteilung, Normalverteilung

(8)

$$\sigma = \sqrt{s^2} = \sqrt{Var[X]}$$

(9)

$$s^2 = \frac{1}{N-1} \sum_{i=1}^{N} (T_i - \bar{T})^2$$

(10)

$$\mu_k = E\left[(X - \mu)^k\right]$$

(11) $Y=(aX+b)^2$ $\qquad X\sim N(0,1)$

$\mu \qquad \sigma = E[X] = 1$

$= \sqrt{Var[X]} = 1$

$Var[X] = 1^2 = 1$

$E[X^2] = 1$

$E[X^4] = 3$

$$
\begin{aligned}
Var[Y] &= Var[(aX+b)^2] \\
&= Var[a^2X^2 + 2abX + b^2] \\
&= Var[a^2X^2] + Var[2abX] \\
&= a^4\, Var[X^2] + 4a^2b^2\, Var[X] \\
&= a^4\, E[(X^2-\mu)^2] + 4a^2b^2 \cdot 1 \\
&= a^4\, E[X^4 - 2\mu X^2 + \mu^2] + 4a^2b^2 \\
&= a^4 \left(E[X^4] - 2\mu E[X^2] + \mu^2\right) + 4a^2b^2 \\
&= a^4(3 - 2\cdot 0 \cdot 1 + 0) + 4a^2b^2 \\
&= \underline{a^2(3a^2 + 4b^2)}
\end{aligned}
$$

(12)

$$
C = \begin{pmatrix} \overset{\sigma_{x_1}^2}{2} & 2 \\ 2 & \underset{\sigma_{x_2}^2}{3} \end{pmatrix}
$$

müssen gleich sein Covar $\to 2$

$\sigma_S = \sqrt{Var[S]}$

$$
\begin{aligned}
Var[S] &= Var[X_1 + X_2] \\
&= Var[X_1] + Var[X_2] + 2 \cdot Covar[X_1, X_2] \\
&= 2 + 3 + 2 \cdot 2 \\
&= 9
\end{aligned}
$$

$\sigma_S = \sqrt{Var[S]}$

$= \sqrt{9} = \underline{3}$

(13) $IQ \sim N(100, 15^2)$

$$P(IQ > 130) = 1 - P(IQ \leq 130) = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right)$$

$$= 1 - \Phi\left(\frac{130-100}{15}\right)$$

$$= 1 - \Phi\left(\cancel{\frac{30}{15}}\,2\right) = 1 - \Phi(2) = 1 - 0{,}9772 = \underline{0{,}0228}$$

(14)

relativ selten → Poisson-Verteilung

$1 \times$ pro Monat

$>5 \times$ pro 3 Monate

$\lambda = 3$ da $1\times$ pro Monat und es 3 Monate sind

$$P(X > 5) = 1 - P(X \leq 5)$$

$$= 1 - e^{-\lambda} \sum_{i=0}^{5} \frac{\lambda^i}{i!}$$

$$= 1 - e^{-3}\left(\frac{3^0}{1} + \frac{3^1}{1} + \frac{3^2}{2} + \frac{3^3}{6} + \frac{3^4}{24} + \frac{3^5}{120}\right)$$

$$= 1 - e^{-3}\left(1 + 3 + \underbrace{\frac{9}{2}}_{4{,}5} + \underbrace{\frac{3\cdot 3\cdot 3}{6}}_{4{,}5} + \underbrace{\frac{3\cdot 3\cdot 3\cdot 3}{24}}_{3{,}375} + \underbrace{\frac{3\cdot 3\cdot 3\cdot 3\cdot 3}{120}}_{2{,}025}\right)$$

$$= 1 - 0{,}916$$

$$= 0{,}084 \rightarrow \underline{8{,}4\%}$$

(15)

$1-\alpha \rightarrow$ Tabelle $(1 - 0{,}05 = 0{,}95 \xrightarrow{\lim \infty} 1{,}96)$

(16)

$N = 25$

$\bar{A} = 50$

$\sigma_A = 30$

$\alpha = 0{,}05$

$$P\left(\bar{A} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{N}} \leq \mu \leq \bar{A} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{N}}\right) = 1 - \alpha$$

$$50 - 1{,}96 \cdot \frac{30}{\underbrace{\sqrt{25}}_{5}} \leq \mu \leq 50 + 1{,}96 \cdot \frac{30}{\sqrt{25}})$$

$$50 - 11{,}76 \leq \mu \leq 50 + 11{,}76)$$

$$\underline{38{,}24} \leq \mu \leq \underline{61{,}76}$$